Database Architectures and the Web

CSE 341
Introduction to Database Systems
Winter 2024
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Objectives

- Client–server architecture
- Difference between two-tier, three-tier and *n*-tier client–server architectures
- Application server
- Middleware

Multi-user DBMS Architectures

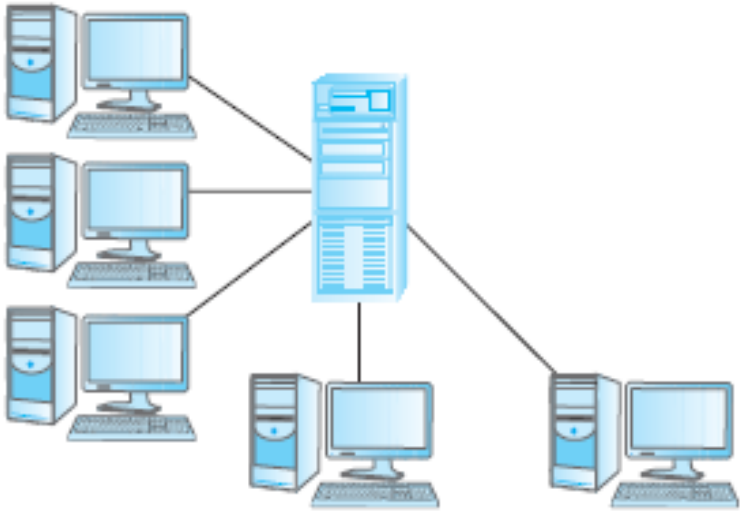
- The common architectures that are used to implement multi-user database management systems:
  - Teleprocessing
  - File-Server
  - Client-Server

File Server

- File-server is connected to several workstations across a network.
- Database resides on file-server.
- DBMS and applications run on each workstation.
- Disadvantages include:
  - Significant network traffic.
  - Copy of DBMS on each workstation.
  - Concurrency, recovery and integrity control more complex.

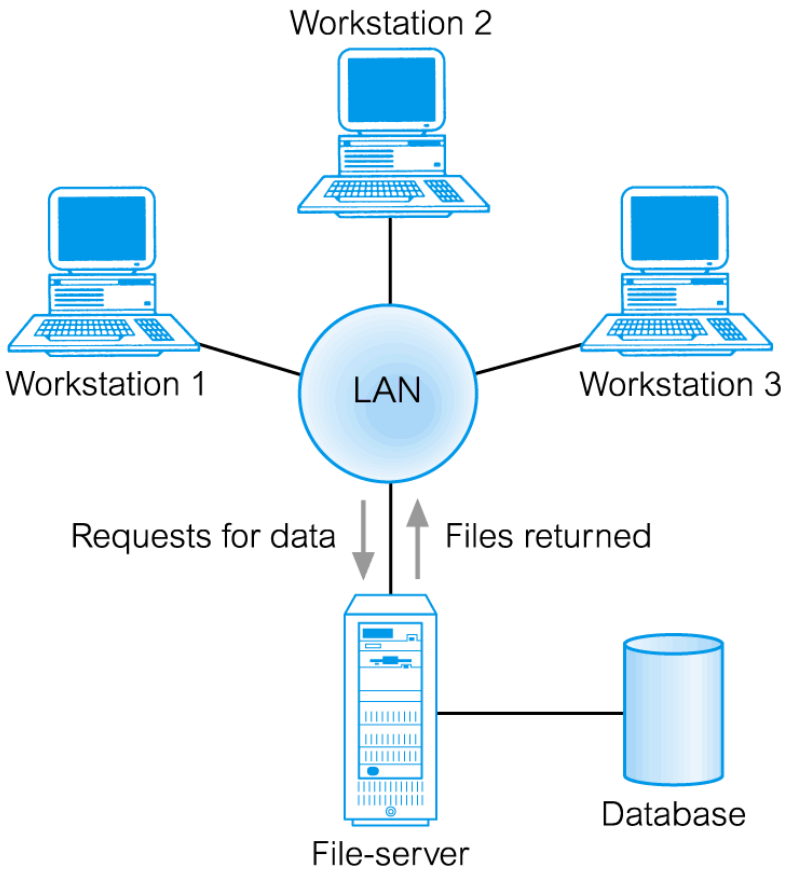
Teleprocessing

- One computer with a single CPU and a number of terminals.
- Processing performed within the same physical computer. User terminals are typically "dumb", incapable of functioning on their own, and cabled to the central computer.



File-Server Architecture

- In a file-server environment, the processing is distributed about the network, typically a local area network (LAN).

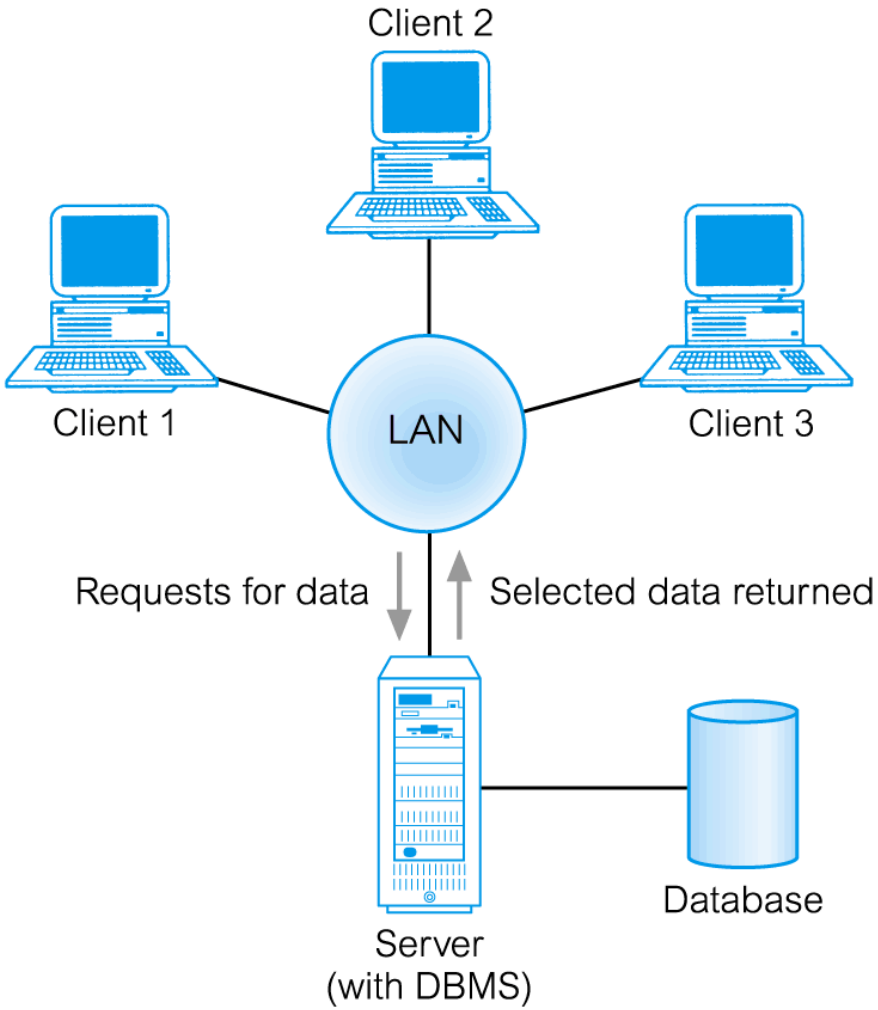


Traditional Two-Tier Client-Server

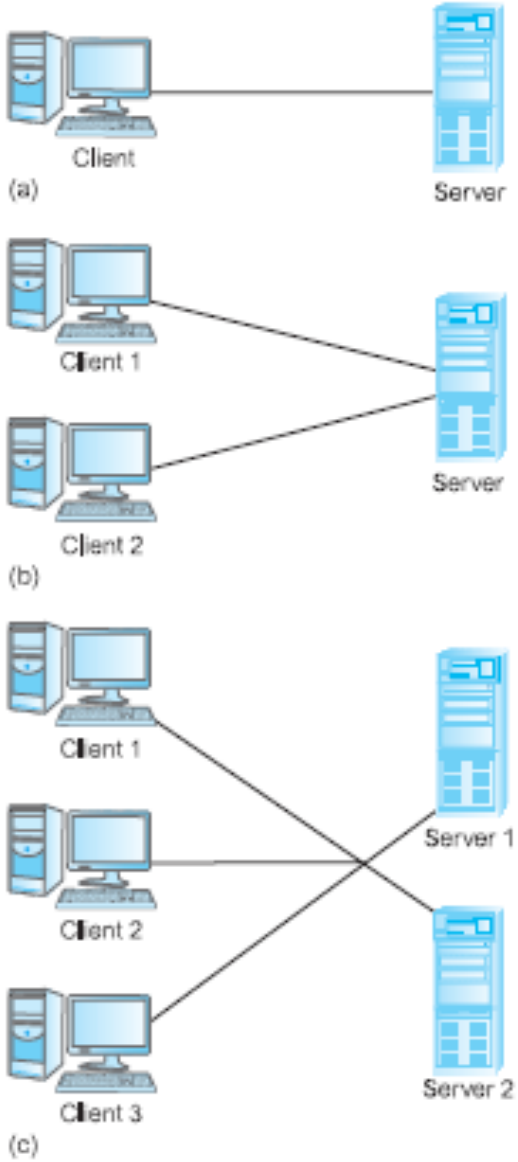
- Client (tier 1) manages user interface and runs applications.
- Server (tier 2) holds database and DBMS.

- Advantages include:
  - wider access to existing databases;
  - increased performance;
  - possible reduction in hardware costs;
  - reduction in communication costs;
  - increased consistency.

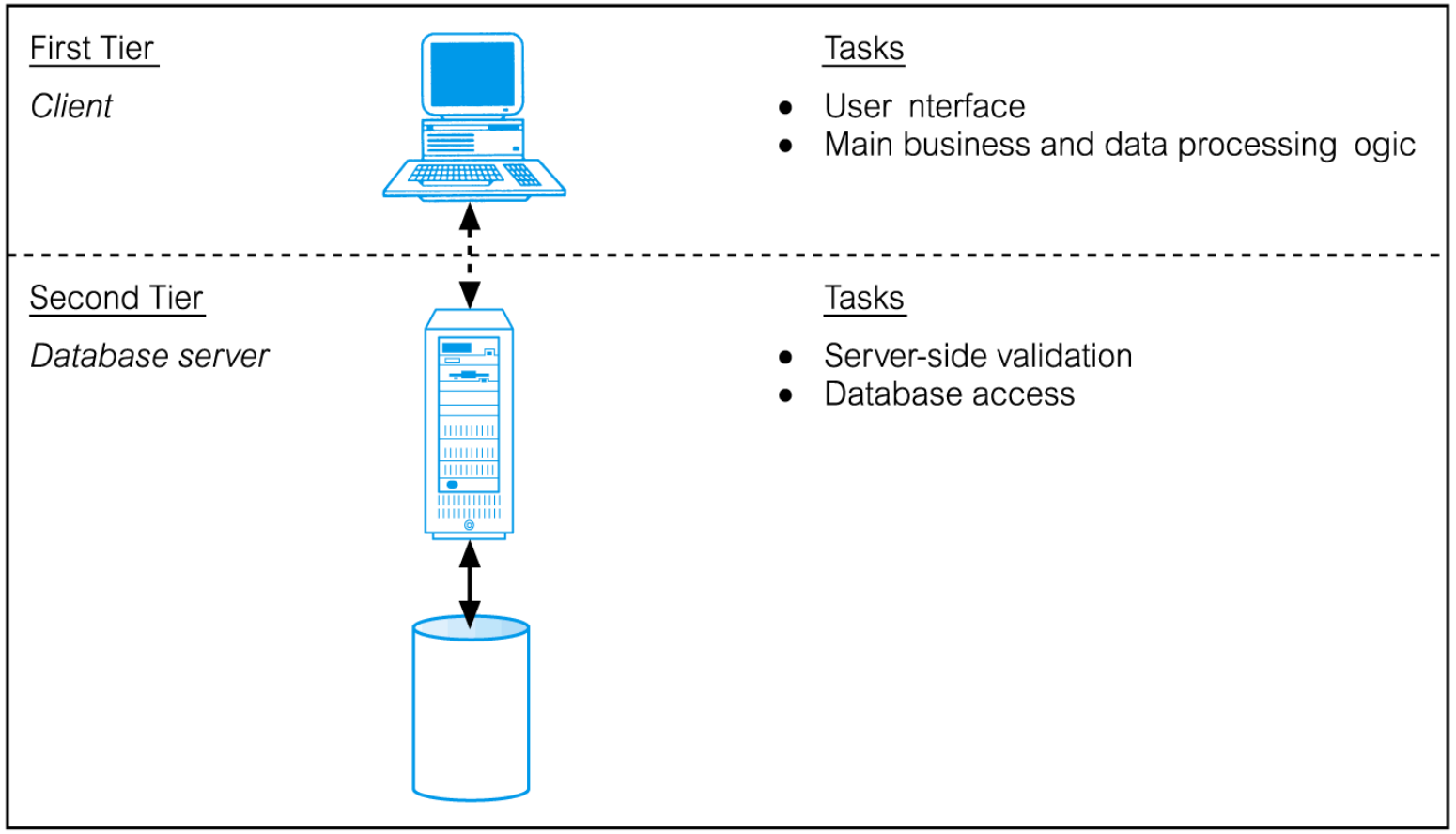
Traditional Two-Tier Client-Server



Alternative Client-Server Topologies



Traditional Two-Tier Client-Server



Summary of Client-Server Functions

CLIENT	SERVER
Manages the user interface	Accepts and processes database requests from clients
Accepts and checks syntax of user input	Checks authorization
Processes application logic	Ensures integrity constraints not violated
Generates database requests and transmits to server	Performs query/update processing and transmits response to client
Passes response back to user	Maintains system catalog Provides concurrent database access Provides recovery control

Three-Tier Client-Server

- The need for enterprise scalability challenged the traditional two-tier client–server model.

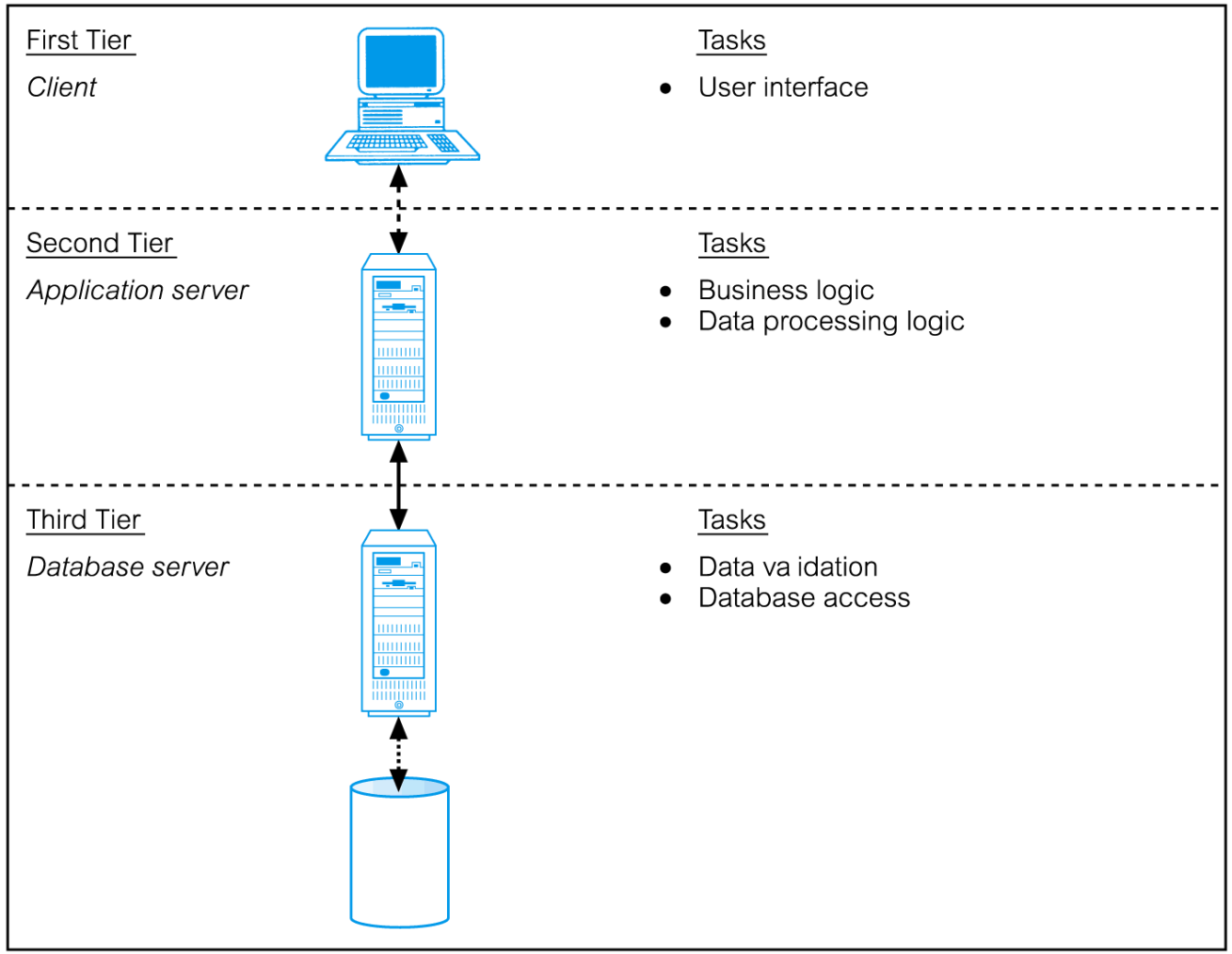
- Client side presented two problems preventing true scalability:
  - 'Fat' client, requiring considerable resources on client's computer to run effectively.
  - Significant client side administration overhead.

- By 1995, three layers proposed, each potentially running on a different platform.

Three-Tier Client-Server

- Advantages:
  - ‘Thin’ client, requiring less expensive hardware.
  - Application maintenance centralized.
  - Easier to modify or replace one tier without affecting others.
  - Separating business logic from database functions makes it easier to implement load balancing.
  - Maps quite naturally to Web environment.

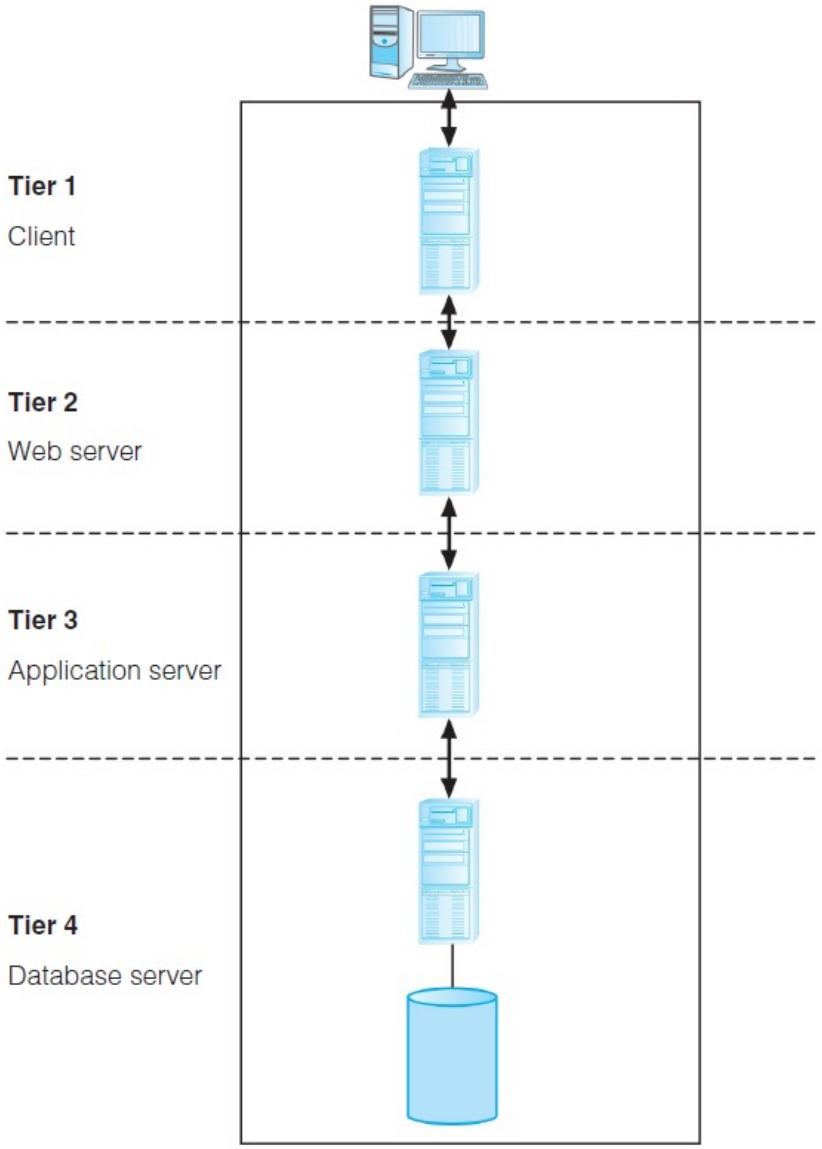
Three-Tier Client-Server



n-Tier Client-Server (e.g. 4-Tier)

- The three-tier architecture can be expanded to *n* tiers, with additional tiers providing more flexibility and scalability.

- Applications servers host API to expose business logic and business processes for use by other applications.



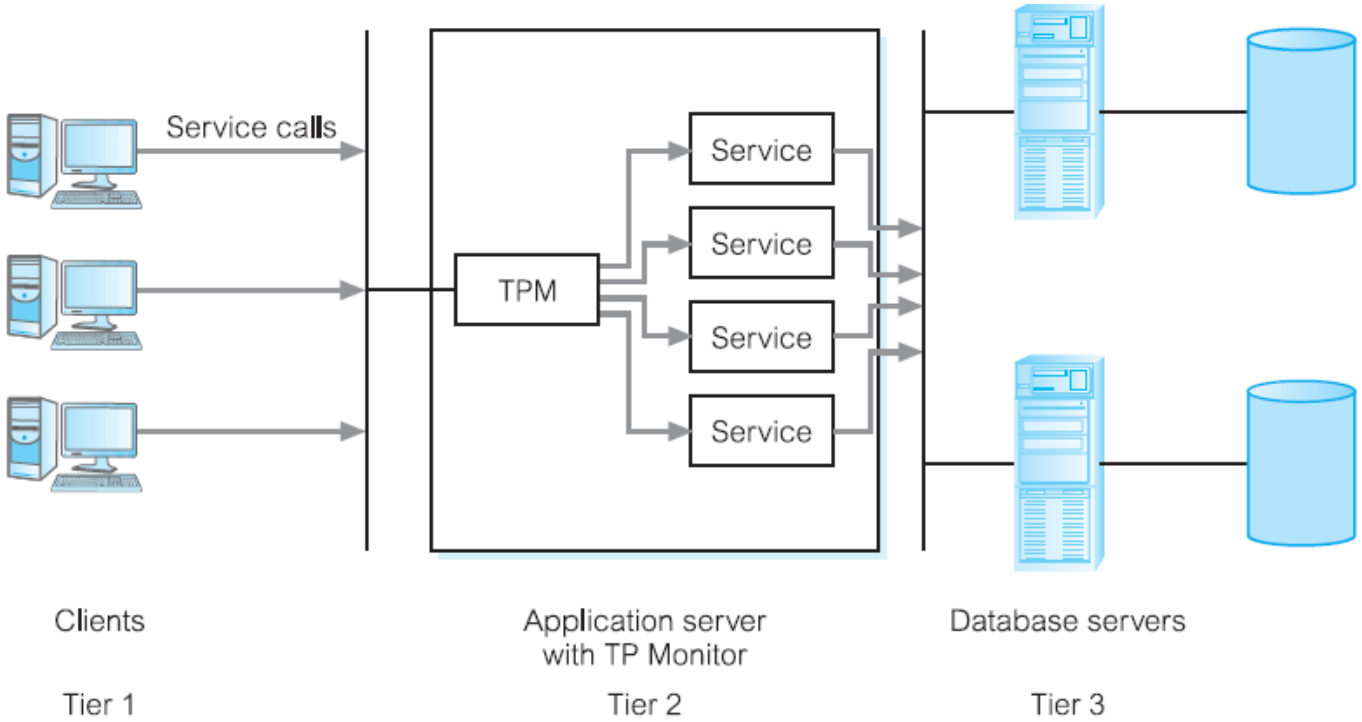
Middleware

- Middleware is a generic term used to describe software that mediates with other software and allows for communication between disparate applications in a heterogeneous system.

- The need for middleware arises when distributed systems become too complex to manage efficiently without a common interface.

Transaction Processing Monitors

- TP monitor is a middleware program that controls data transfer between clients and servers in order to provide a consistent environment, particularly for online transaction processing (OLTP).



Web Services and Service-Oriented Architectures

- Web service is a software system designed to support interoperable machine-to-web service machine interaction over a network.

- Unlike other Web-based applications, Web services have no user interface and are not aimed at Web browsers.

- Web services share business logic, data, and processes through a programmatic interface across a network.

Web Services and Service-Oriented Architectures

Examples of Web services include:

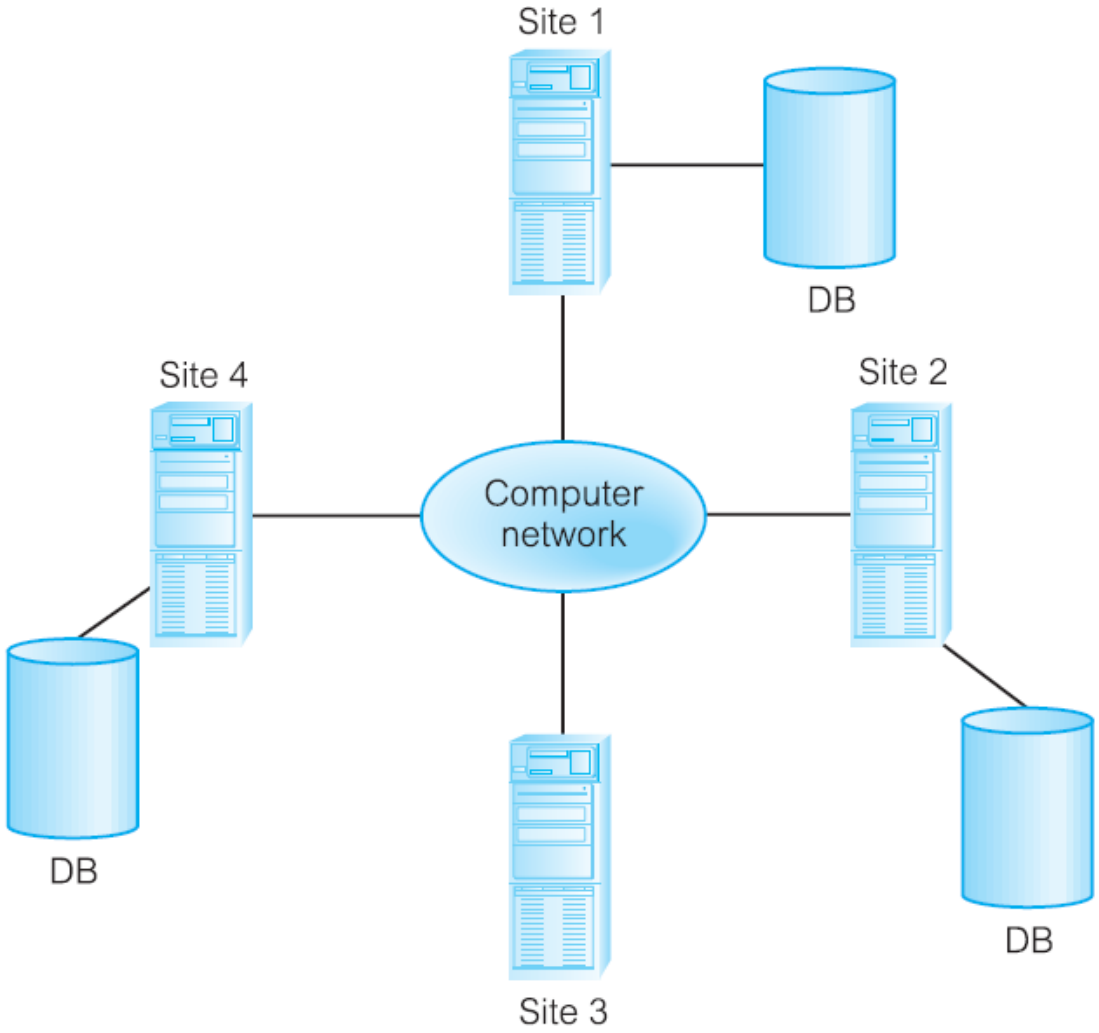
- Microsoft Bing Maps and Google Maps Web services provide access to location-based services, such as maps, driving directions, proximity searches, and geocoding (that is, converting addresses into geographic coordinates) and reverse geocoding.
- Amazon Simple Storage Service (Amazon S3) is a simple Web services interface that can be used to store and retrieve large amounts of data, at any time, from anywhere on the Web. It gives any developer access to the same highly scalable, reliable, fast, inexpensive data storage infrastructure that Amazon uses to run its own global network of Web sites

Distributed DBMSs

- Distributed Database: A logically interrelated collection of shared data (and a description of this data), physically distributed over a computer network.

- Distributed DBMSs: The software system that permits the management of the distributed database and makes the distribution transparent to users.

Distributed DBMSs



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Entity-Relationship Modeling (ER)